GLENCORE

Maintenance Work Management Guideline

5. WORK CLOSED OUT

08

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Comment/s	Update to Measures of Success

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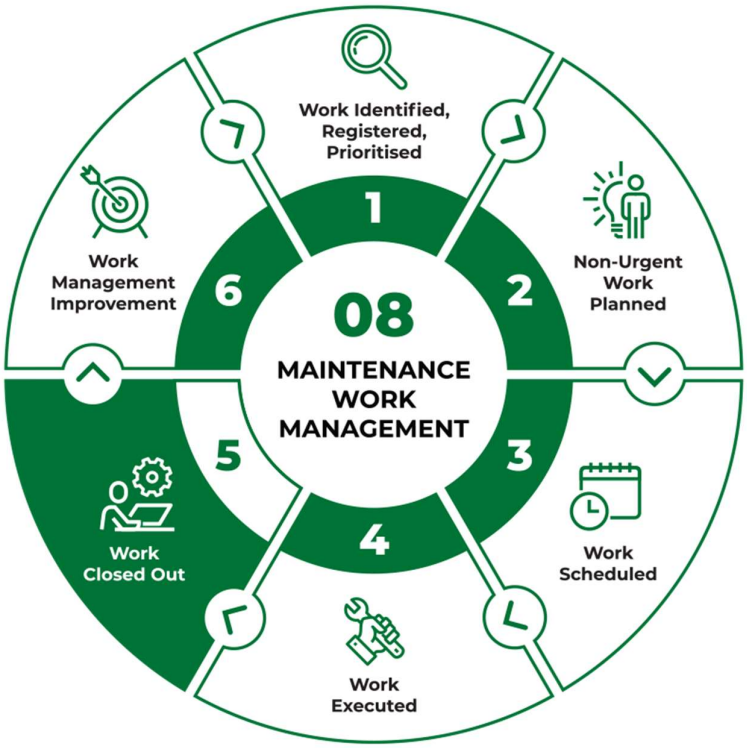
Sub-Process Overview

Purpose

The Glencore Copper Asset Management Framework (AMF) enables the business to deliver its Asset Management purpose and vision, as defined in the AMF Policy. The Framework consists of 18 Elements, detailed in Manuals that define how Glencore Copper manages the whole life of its assets to optimise value and ensure assets are capable of meeting the business and stakeholder objectives.

Maintenance Work Management is Element 08 of the AMF and defines effective work management practices and performance requirements to enable work to be performed safely, efficiently, with precision and when required.

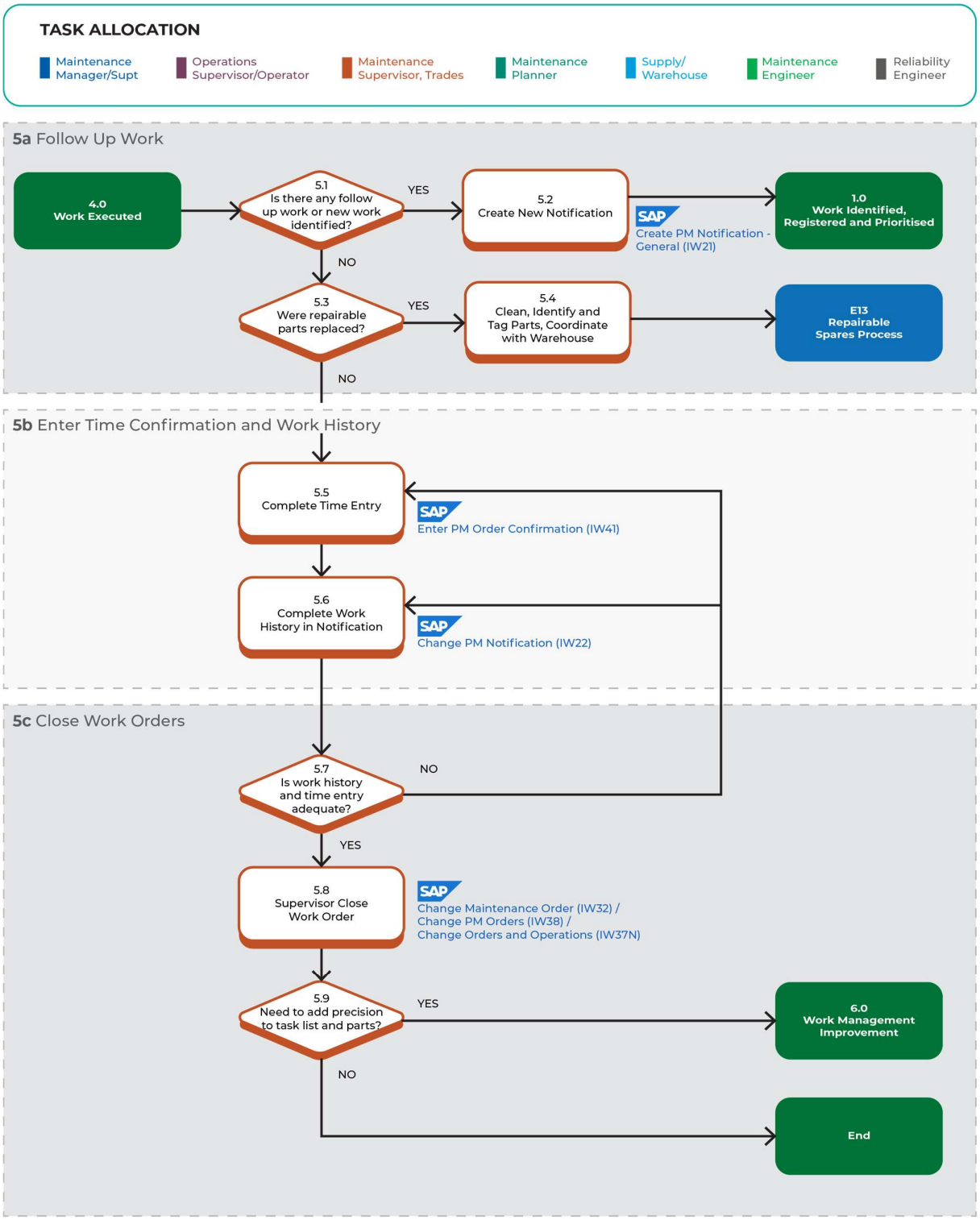
This document is intended to provide end users with understanding and knowledge of sub-process **5. Work Closed Out.** This includes how actual work time and work history are recorded in Work Orders and Notifications, and how they are closed.



Objectives & Flowchart

The objective of sub-process **5. Work Closed Out** is to:

- Identify any follow up work, and ensure that deferred work gets rescheduled.
- Manage any Rotables (repairable items) that were removed.
- Record actual work times (time confirmation) and actual work history onto the Work Order and Notification.
- Record equipment failure data such as Damage and Cause codes.
- Close the Notification and Technically Complete (TECO) the work order.
- Add counters or meters data (hour meters, odometer readings etc.).
- Complete Service Entry Sheets (SES) for site contract work.


TASK ALLOCATION
Maintenance Manager/Supt
Operations Supervisor/Operator
Maintenance Supervisor, Trades
Maintenance Planner
Supply/ Warehouse
Maintenance Engineer
Reliability Engineer
5a Follow Up Work
4.0 Work Executed
5.1 Is there any follow up work or new work identified?
YES
5.2 Create New Notification
SAP Create PM Notification - General (IW21)
1.0 Work Identified, Registered and Prioritised
NO
5.3 Were repairable parts replaced?
YES
5.4 Clean, Identify and Tag Parts, Coordinate with Warehouse
E13 Repairable Spares Process
NO
5b Enter Time Confirmation and Work History
5.5 Complete Time Entry
SAP Enter PM Order Confirmation (IW41)
5.6 Complete Work History in Notification
SAP Change PM Notification (IW22)
5c Close Work Orders
5.7 Is work history and time entry adequate?
NO
YES
5.8 Supervisor Close Work Order
SAP Change Maintenance Order (IW32) / Change PM Orders (IW38) / Change Orders and Operations (IW37N)
5.9 Need to add precision to task list and parts?
YES
6.0 Work Management Improvement
NO
End

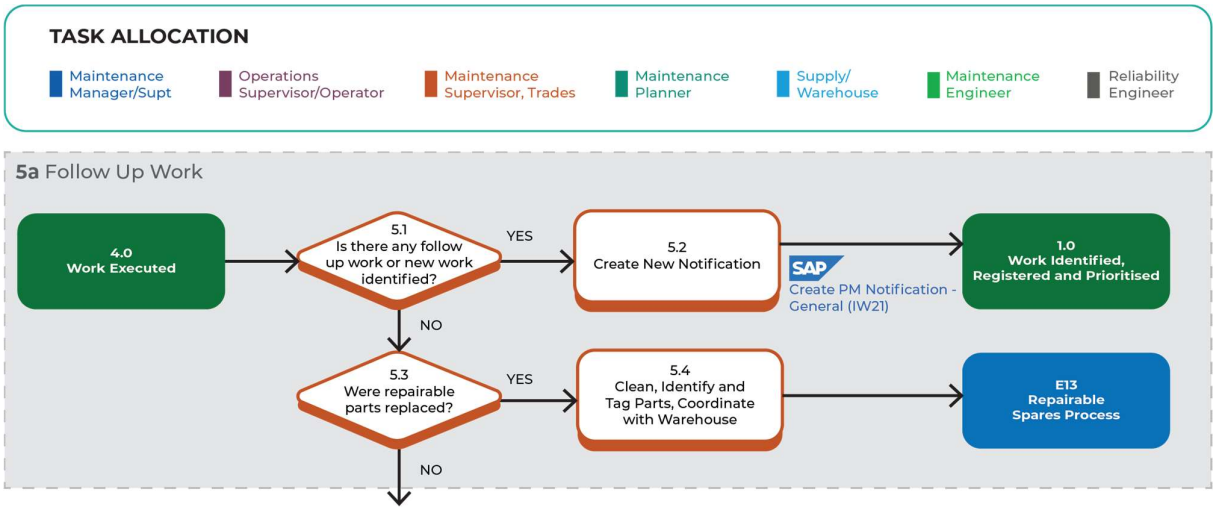
Roles & Responsibilities

R Responsible
S Support
I Inform

		Operations			Maintenance				Other		
		Superintendent	Supervisor	Operator	Superintendent	Supervisor	Trades	Planner	Maintenance Engineer	Reliability Engineer	Supply / Warehouse
5.1	Is there any follow up work or new work identified?					S	R				
5.2	Create new Notification						R				
5.3	Were Repairable parts replaced?						R				
5.4	Clean, identify and tag repairable parts, coordinate with warehouse					R	S				S
5.5	Complete Time Entry						R				
5.6	Complete work history in Notification						R				
5.7	Is work history and time entry adequate?					R					
5.8	Close WO					R					
5.9	Need to add precision to task list and parts?					R					

5. Work Close Out

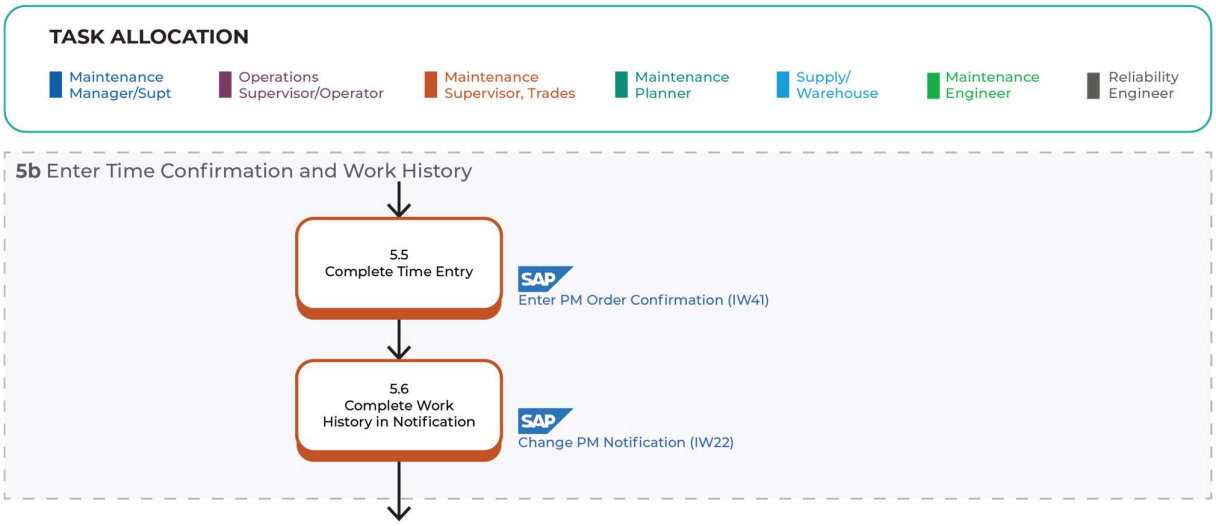
5a Follow Up Work



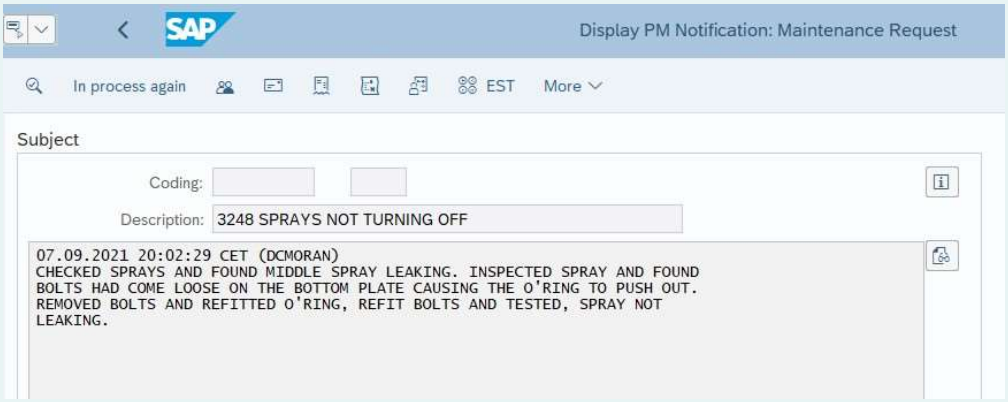
What?	Create new Notifications for any follow up (or new) work and process any repairable equipment (rotables) from completed work.
Why?	Any follow up work must be raised as a Subsequent Notification so that it is captured and can get planned, scheduled and executed. This is important to demonstrate where defects have been identified when executing PM02 inspections and services, highlighting the benefits of proactive maintenance and any strategy improvement opportunities. Repairable items (Rotables) removed during work execution must be managed according to a specific process, to ensure that they get repaired and returned to stores.
Who?	• Tradespersons • Maintenance Supervisor *Note: Maintenance Supervisor may delegate responsibility for processing Rotables to a Leading Hand / Technical Officer.*
How?	For any job that requires follow up work (e.g. a temporary fix was completed but a permanent repair is still needed) or where additional work was completed or identified (e.g. as part of a PM02 inspection or service), raise a Subsequent Notification in SAP using **Create PM Notification – General (IW21)** to capture this information and link to the original Notification. Refer to **SAP PM QRG – Create Subsequent Notification** for instructions. Refer to **1b Create Notifications in SAP** in **1. Work Identified, Registered & Prioritised** guideline document for Notification requirements. Any Rotable item removed during work must be: • Returned to the workshop • Cleaned • Tagged according to local site procedure for Rotables • Delivered to the designated Rotables drop-off location at the Warehouse

	Refer to **SAP PM QRG - Add Rotable Component for Change Out** for detail on SAP transactions relating to rotable components. Return any unused parts and materials to the stores. Refer to **E13 Procurement and Supply Chain Management** for relevant process and procedure information.
When?	Daily

5b Enter Time Confirmation and Work History



What?	Enter Time Confirmations and Work History into SAP.
Why?	Entering the actual time taken to execute Work Orders is how costs are allocated to the relevant equipment and is also used to assess the accuracy of planning. Adding work history and other comments provides an important record of what work was actually done to the equipment and is critical for root cause, reliability, and defect elimination studies along with repair/replace decision making.
Who?	• Tradespersons *Note: In some work areas, data entry of time confirmations and work history from Tradespersons (handwritten on job sheets) into SAP may be delegated to an administrative clerk.*
How?	Tradespersons must complete Time Confirmation in SAP using **Enter PM Order Confirmations (IW41)** (refer to **SAP PM QRG – Time Confirmation - Notification Completion** for details) as follows: • Enter actual work hours taken to execute the work, for every relevant Work Order and Operation they worked on. • Enter the estimated time in the remaining work field where applicable. • Enter the reason for any job delays. • When no more time is to be added, tick “Final Confirmation”. This step informs the Maintenance Supervisor that the Work Order or Operation is completed. Tradespersons must then enter Work History in the Notification in SAP either via navigating to the Notification via the “Notification” button in **Enter PM Order Confirmations (IW41)** or by using **Change PM Notification (IW22)** (also refer to **SAP PM QRG – Time Confirmation - Notification Completion** for details) as follows: • Enter relevant work history notes into the subject long text box. ○ History should include a description of what work was actually completed, the root cause of the defect (if known), and any other relevant notes such parts and materials, operating parameters etc. Refer below for an example of good work history notes:



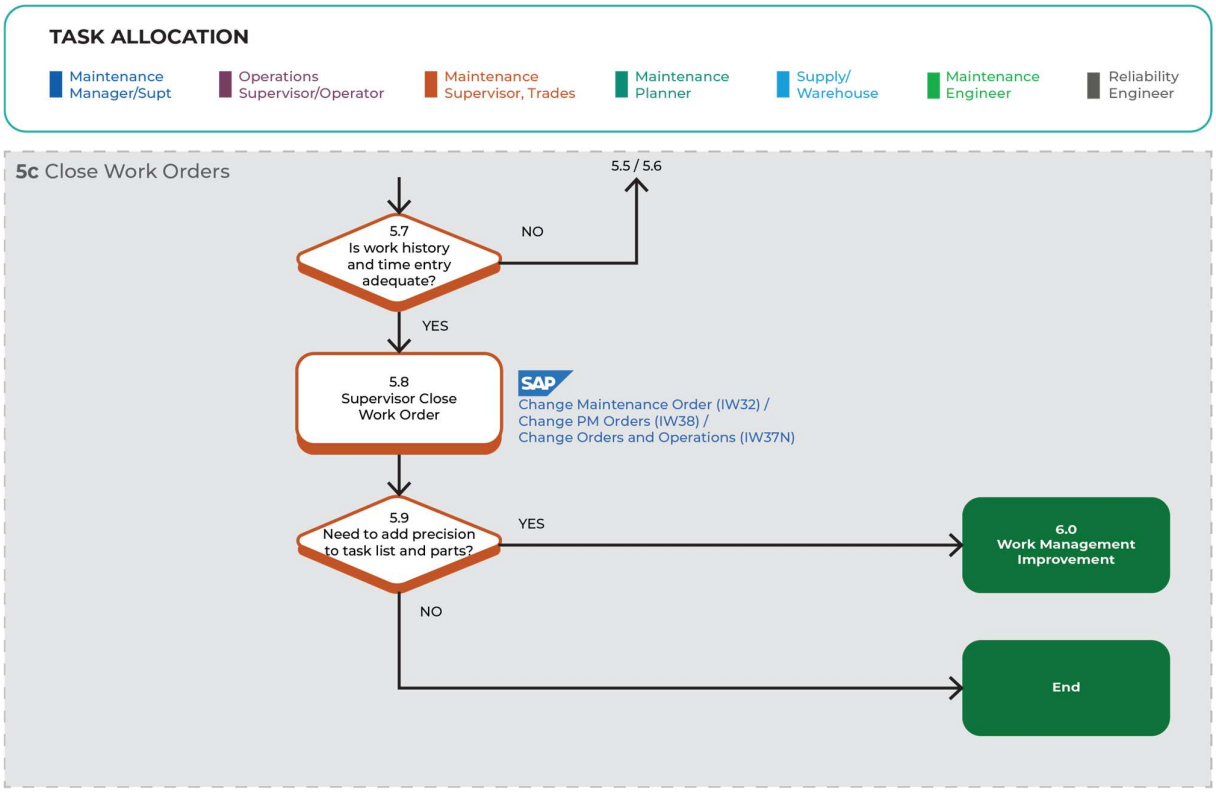
- Confirm (or update) the Cause, Damage and Object codes.
- Complete the Notification by selecting the "Complete" flag to apply the Notification system status Notification Complete (NOCO).

Upon completion of Time Confirmations and Work History, return any hard copy documents (Work Orders, inspection sheets, service sheets etc.) to the Maintenance Supervisor for record keeping.

Where data entry of Time Confirmations and Work History from Tradespersons are entered into SAP by an administrative clerk, refer to **SAP PM QRG – Collective Time Confirmations (Bulk)**.

When? Daily

5c Close Work Orders



What?	Check that the work has been executed and that time confirmations and work history have been entered, before making the Work Orders Technically Complete (TECO). Change Work Order status for any deferred work. Identify any improvement opportunities.
Why?	Once work has been executed, the relevant Work Order is ready to be closed (TECO). However, the Maintenance Supervisor must conduct a series of check to ensure that all necessary steps have been performed before they can close the Work Order. Work not started must be deferred so it can be rescheduled. Capturing any potential improvement opportunities (i.e. identified by the Trades) contributes to the continuous improvement and optimisation cycle detailed in **6. Work Management Improvement**.
Who?	• Maintenance Supervisor
How?	All Work Orders that have been executed in the period must be checked and closed (TECO) according to the following steps. In SAP using **Change PM Orders (IW38)**, generate a list of work orders for the workgroup ready to close out by searching for System Status CNF (Confirmed). Alternatively, use **Change Orders and Operations (IW37N)** to show Work Order Operations and the status codes against each. Check the following (referring to **SAP PM QRG – Technically Complete (TECO) PM Order**): • Time Confirmations have been entered against all Operations.

- Work History has been entered, and Object, Damage and Cause codes entered (note that TECO will not be possible unless the codes have been entered).
- All parts and materials have been issued, including all negative reservation (i.e. returns to stores including Rotables).
- Check actual versus planned costs.
- Tick the Technically Complete (TECO) Flag.
    - For breakdown Work Orders (where the breakdown box was ticked in the Notification), you will be prompted to enter the Malfunction End date and time.

Note: If the Work Order is TECO'd before materials have been issued, all reservations will be deleted. If all maintenance activities for the WO have been completed, but waiting on reports (e.g. statutory and critical control work), assign the user status Maintenance Complete (MCMP). The Maintenance Planner should complete a weekly search for Work Orders with MCMP status, excluding TECO, and TECO where ready.

For PM02 Work Orders using a counter driven strategy (e.g. equipment hours, odometer readings), after TECO, go to the Create Measurement Document tab (**Create Measurement Document (IK11)**) and enter the new Counter reading.

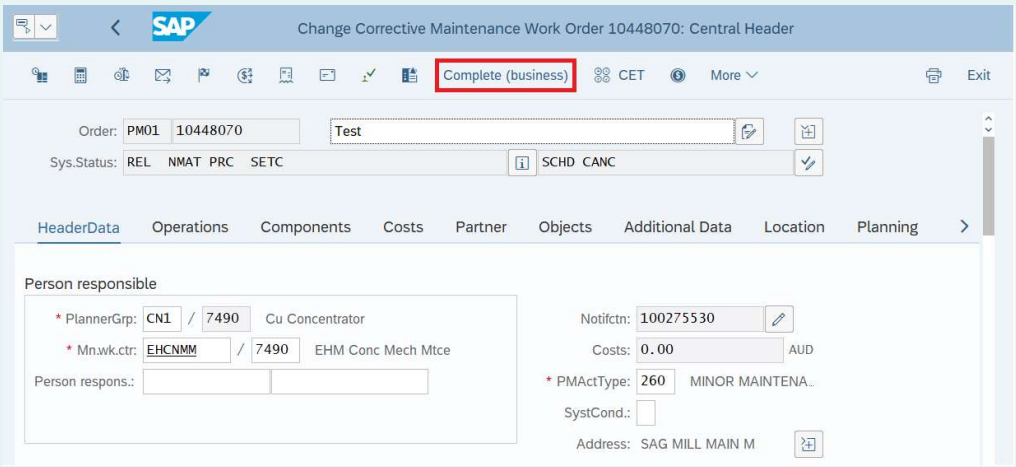
For all completed Contractor services, enter the Service Entry Sheet (SES) into SAP using **Service Entry Sheet (ML81N)** (refer to **SAP PM QRG - Service Entry Sheet** for further details). The Service Entry Sheet is the record of what actual hours the contractor spent on site, which will later be reconciled against the Purchase Order and the Invoice by the purchasing group.

Note: SES's should be entered before TECO of a Work Order, but can be completed after if necessary.

Where work was not completed, select the Work Order User Status Rescheduled (RSCH) and an appropriate reason (User Status without Number) as per the following table:

User Status	Description	Definition
NCNE	Not Completed – Equipment not available	Work not completed because the equipment was not available for maintenance.
NCNM	Not Completed – Material not available	Work not completed because required materials were not available
NCNT	Not Completed – Not enough time	Was not completed because due to lack of available resources (labour).

Where work was not completed and Maintenance will NOT perform the work at a later date, use the "Complete (business)" function in **Change Maintenance Order (IW32)** as shown below. Following that step, select the Work Order user status Cancelled (CANC).



Identify Work Management Improvement Opportunities:

Based on the Time Confirmation or Work History comments, determine whether there is opportunity to improve the future planning of work. Improvements could include:

- Amended Bill of Materials (BOM) and/or parts lists.
- Updates to Work Instructions and/or Task Lists (added job steps, improved details or instructions etc).
- Process failings and inefficiencies (e.g. kitting of parts not performed, job packs missing, equipment isolation delays, equipment not cleaned, permit delays etc.)
- Amended job duration estimates.
- Possible changes / requests for investigation into maintenance strategies.

Where such improvements can be made, make note of them in preparation for sub-process **6. Work Management Improvement** (refer to Element 08 Guideline for details).

When? Daily

Measures of Success

The following are measures of success for sub-process **5. Work Closed Out**, as described in the Glencore Copper Asset Management Manual **Maintenance Work Management** (Element 08).

% Work Orders with all the Work Orders TECO'd within 45 Days	
KPI Number	KPIGPM009
Definition	Measures the percentage of completed (MCMP) work order operations that have been TECO'd (System status is set to TECO) within 45 days of the Basic Finish Date.
Performance Indicator	100% of all work orders TECO'd within 45 days of work execution.
Calculation	$\frac{\text{Work Orders TECO'd within 45 days of completion date}}{\text{Total number of work orders in scheduled period with status change to TECO}} \times 100$
Rationale	The rationale for this metric is to follow orders through to final business completion i.e. status of CLSD.

Planning Accuracy – Overall Cost (%)	
KPI Number	KPIGPM010
Definition	Actual costs (labour and materials) confirmed on completed work orders with a status of MCMP vs the planned costs for the same work orders.
Performance Indicator	80 – 120%
Calculation	$\frac{\text{Total actual costs on work orders with status MCMP (Labour and Materials)}}{\text{Total estimated costs for the same planned work orders}} \times 100$
Rationale	This KPI is important for monitoring the accuracy of the planning function. It compares the actual costs of labour and materials against the estimate for the same when planned.

Planning Accuracy – Overall Hours (%)	
KPI Number	KPIGPM011
Definition	Actual time confirmed hours on completed work orders with a status of MCMP vs the planned labour provision for the same work orders.
Performance Indicator	80 – 120%
Calculation	$\frac{\text{Total time confirmed hours on completed work orders with a MCMP status}}{\text{Total estimated labour hours for the same planned work orders}} \times 100$
Rationale	This KPI is important for monitoring the accuracy of the planning function. It compares the actual time confirmed labour against the estimate for the same when planned.



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